# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 9 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 13: **Earth as a System: Energy, Matter and Life** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** A surface that reflects 40 per cent and absorbs the rest has an albedo of: 1

**(a)** 4 **(b)** 0.6 **(c)** 60 **(d)** 0.4

**2.** If 1 kWh of electricity produces 0.8 kg of CO2, the saving of 2.55 kWh avoids: 1

**(a)** 2.04 kg of CO2 **(b)** 20 kg **(c)** 0.8 kg **(d)** 3.2 kg

**3.** Clouds are formed when moist air: 1

**(a)** rises, cools and condenses **(b)** sinks and warms

**(c)** freezes only **(d)** evaporates

**4.** Denitrifying bacteria convert nitrates to: 1

**(a)** carbon dioxide **(b)** nitrogen gas **(c)** ammonia **(d)** proteins

**5.** Using a cloth bag instead of a plastic bag: 1

**(a)** reduces plastic waste **(b)** increases waste

**(c)** wastes water **(d)** causes pollution

**6.** Solar radiation at the poles is less than at the equator because of: 1

**(a)** ozone **(b)** distance from the Sun

**(c)** the Moon **(d)** slanting rays

**7.** The CO2 content of the air is about 0.04 per cent. In 1 litre of air the volume of CO2 is about: 1

**(a)** 40 mL **(b)** 0.04 mL **(c)** 4 mL **(d)** 0.4 mL

**8.** A large portion of the Earth's fresh water is stored in: 1

**(a)** lakes **(b)** atmosphere

**(c)** rivers **(d)** glaciers and ice caps

**9.** The excess use of nitrogen fertilisers causes: 1

**(a)** more rain **(b)** increase in ozone

**(c)** pollution of water bodies **(d)** reduced CO2

**10.** Switching off the devices from the plug when not in use: 1

**(a)** saves standby energy **(b)** wastes energy

**(c)** damages devices **(d)** increases emissions

## SECTION — B (Very Short Answer Type) 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** State why rainwater is naturally slightly acidic. 2

**12.** Why is ground-level ozone harmful while stratospheric ozone is beneficial? 2

**13.** How does burning of 1 litre of petrol contribute to global warming? Name the gas and the amount (about 2.3 kg of $CO_2$). 2

**14.** Why is the Earth called a system? 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** Describe the effects of acid rain on buildings, soil and aquatic life with the cause. 3

**16.** Explain how the clouds can both cool and warm the Earth. 3

**17.** State the role of the soil in the Earth system and the threats to it. 3

OR

Explain the importance of the biogeochemical cycles for sustaining life. 3

**18.** Compare renewable and non-renewable sources of energy with two examples each. 3

OR

Explain how personal choices (food, transport, energy) change the carbon footprint, with an example of each. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** Describe the effects of CFCs and the Montreal Protocol, and the evidence for the recovery of the ozone layer, with the reasons why the problem was solved relatively quickly. 5

OR

Discuss the potential of solar and wind power in India with the possible problems and solutions such as storage and grid. 5

**20.** Explain the latitudinal variation of solar heating, the angle of incidence of the rays and the atmosphere's thickness, with the comparison of the equator, mid-latitudes and the poles. 5

OR

Explain how the interaction of the four spheres can be illustrated by a volcanic eruption. 5

## END OF QUESTION PAPER — Total: 40 Marks

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